

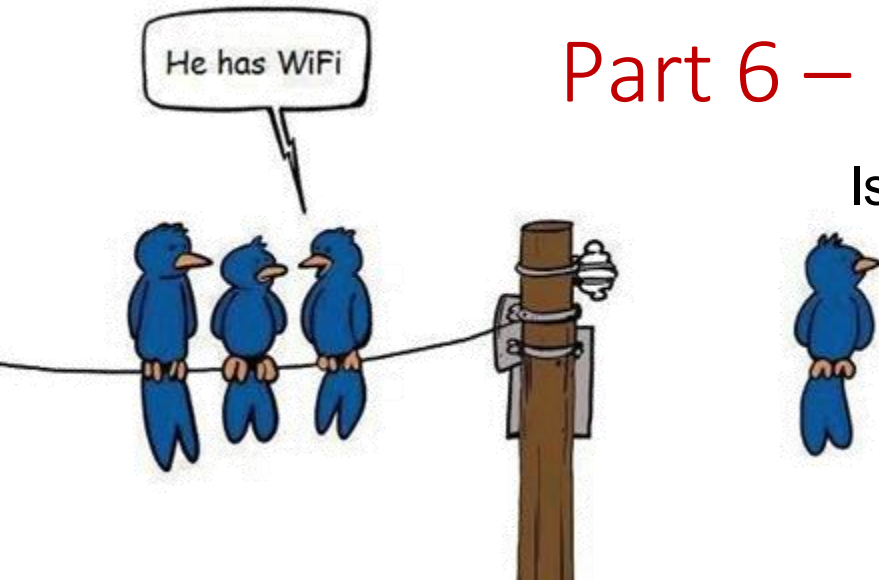
ECSE 4560/ 6560

Modern Communication Systems

Lecture 7: Wireless Systems Overview

Part 6 – Multipath Channel

Ish Jain, Fall 2025



Announcements

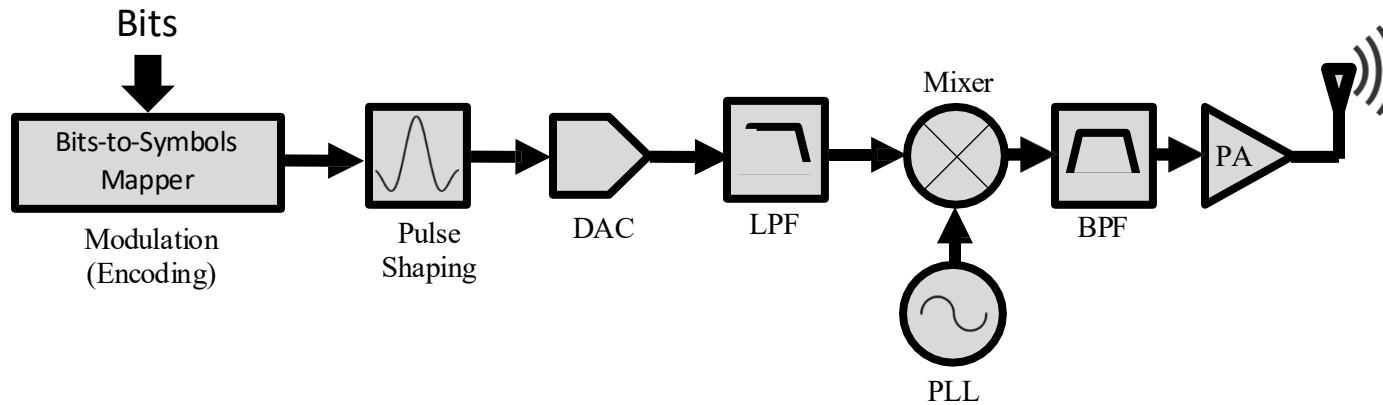
- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **Multipath**
- Projects
 - A list of papers will be posted
 - Group formation [MCS Fall25 project.xlsx](#)
 - Proposal due extended **10/3**
- HW1/Lab1 due Monday 9/22 (extended Wed 9/24)
 - Submit on Gradescope – Mark pages for each question parts
 - Matlab coding exercises – “Use Publish option to submit your code”
- HW2/Lab2 posted due Mon 9/29 (extended Wed 10/1)

Today's Class: Wireless Overview 6

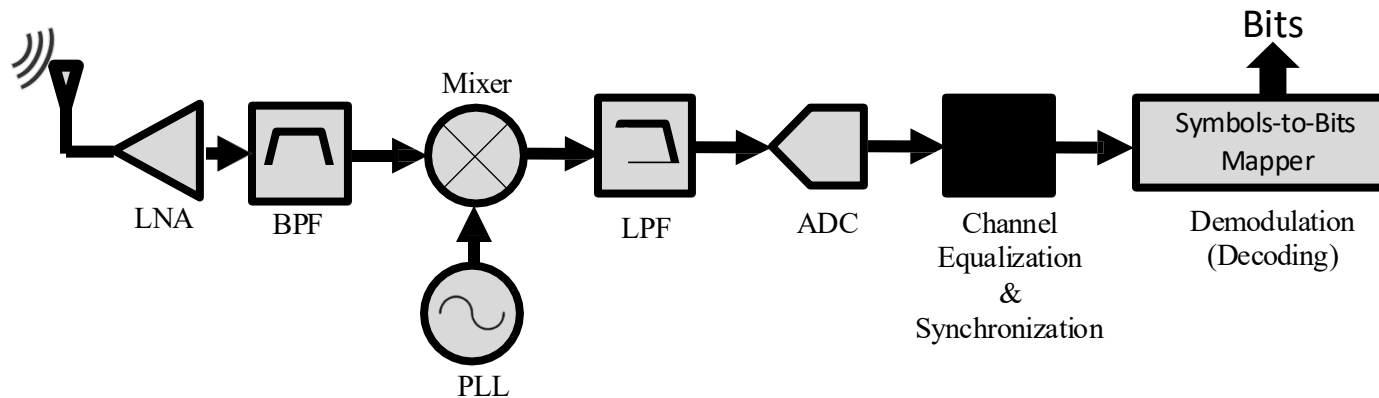
- HW1 and HW2
- Wireless Channel
 - Baseband and passband
- Multipath Channel
 - Power-delay profile
 - Inter-symbol interference
 - Frequency selective fading
 - Narrowband and wideband channel

Wireless Communication System

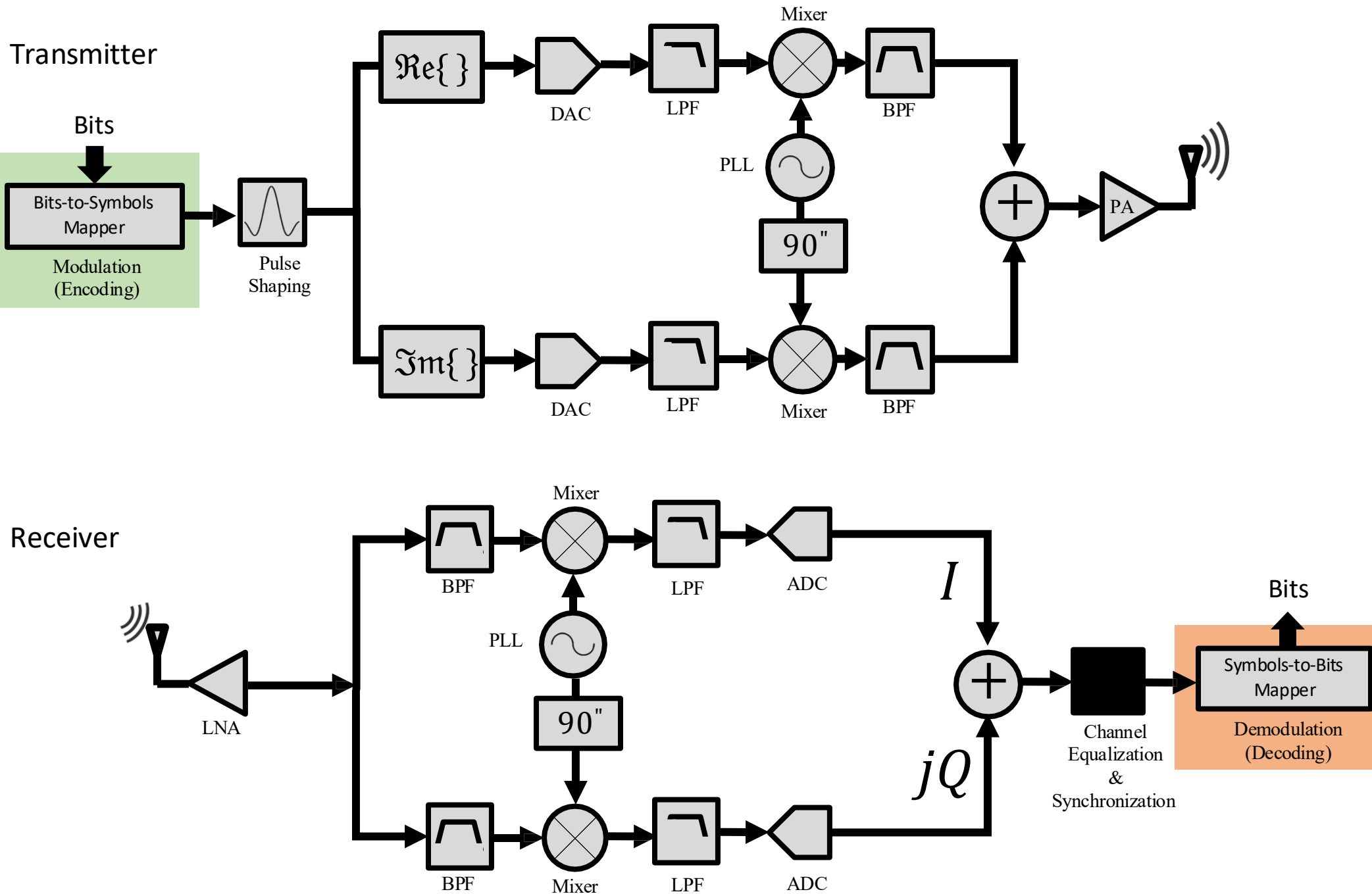
Transmitter



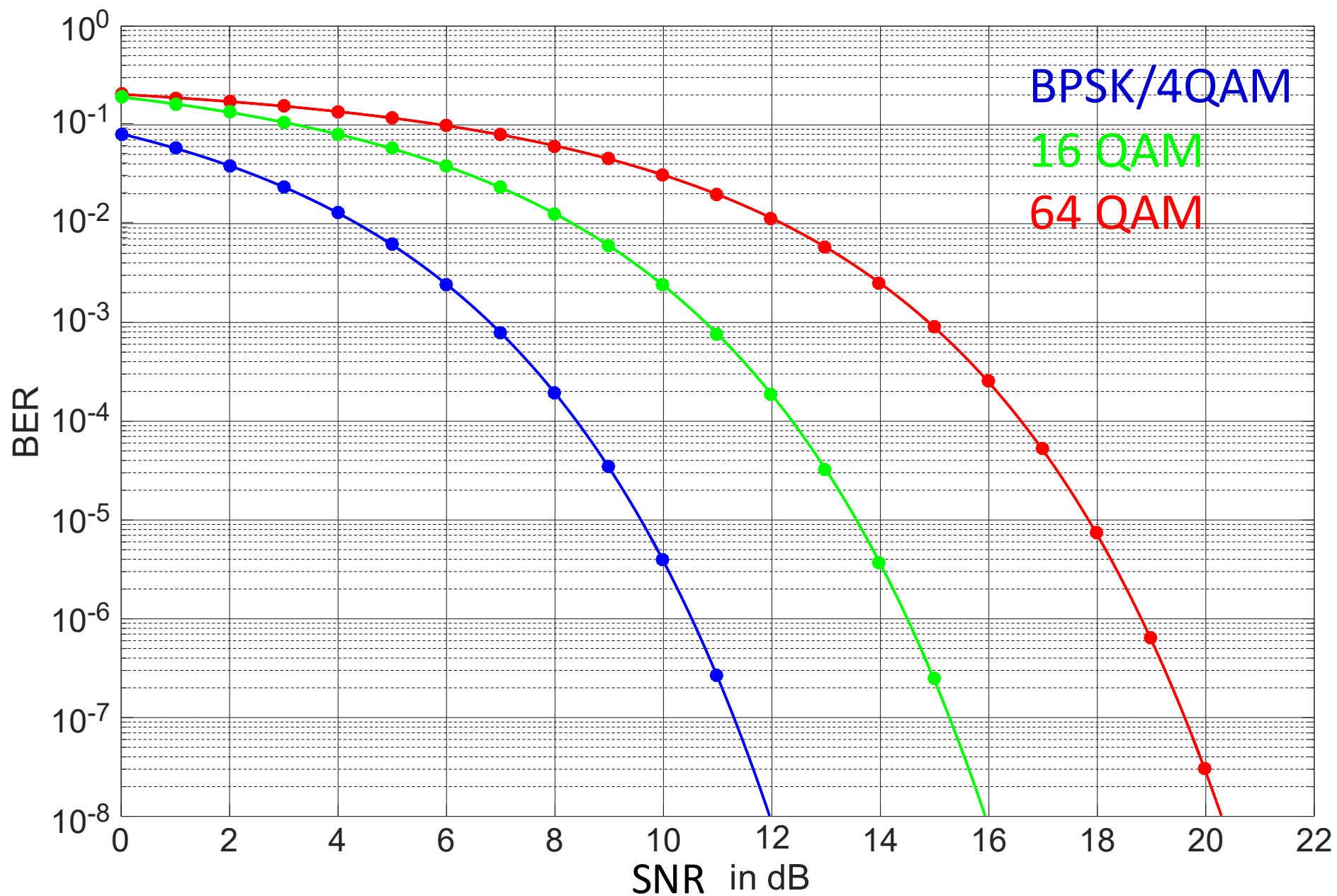
Receiver



Wireless Communication System



Bit Error Rate (BER)

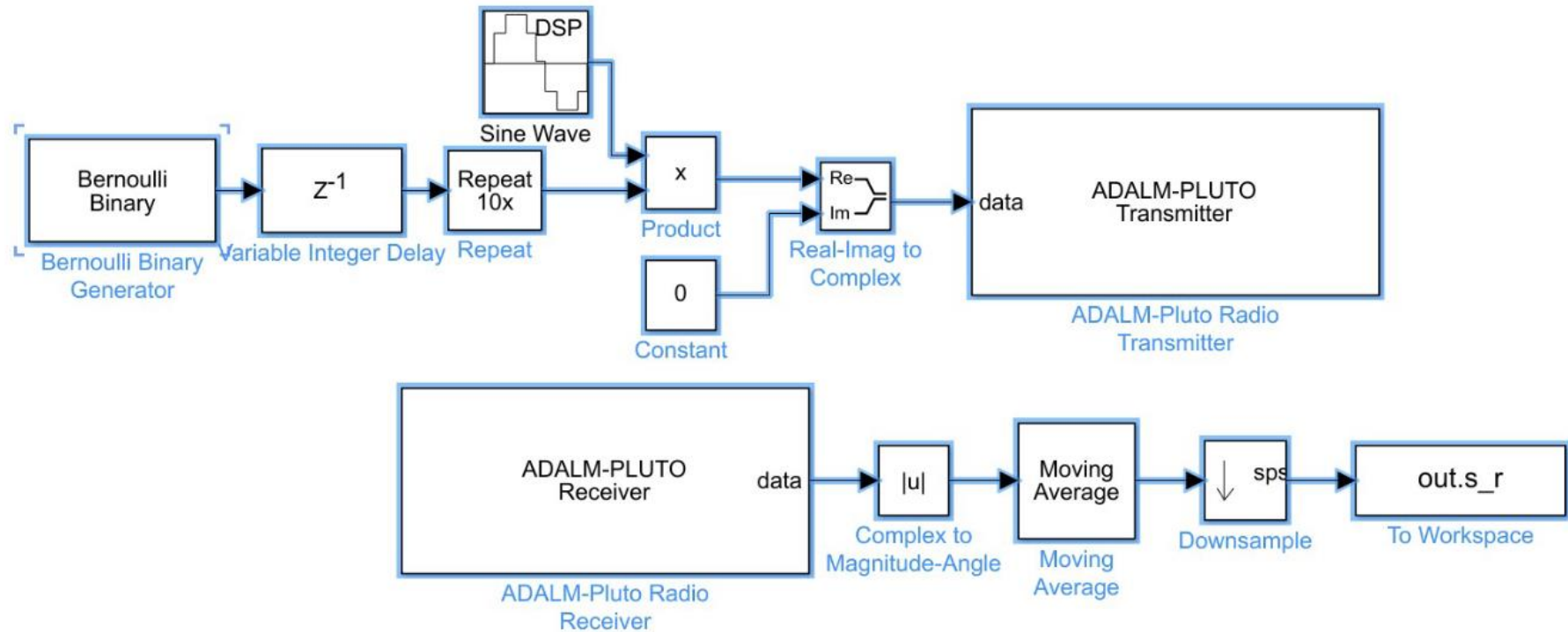


Sampling gain to improve eff. SNR

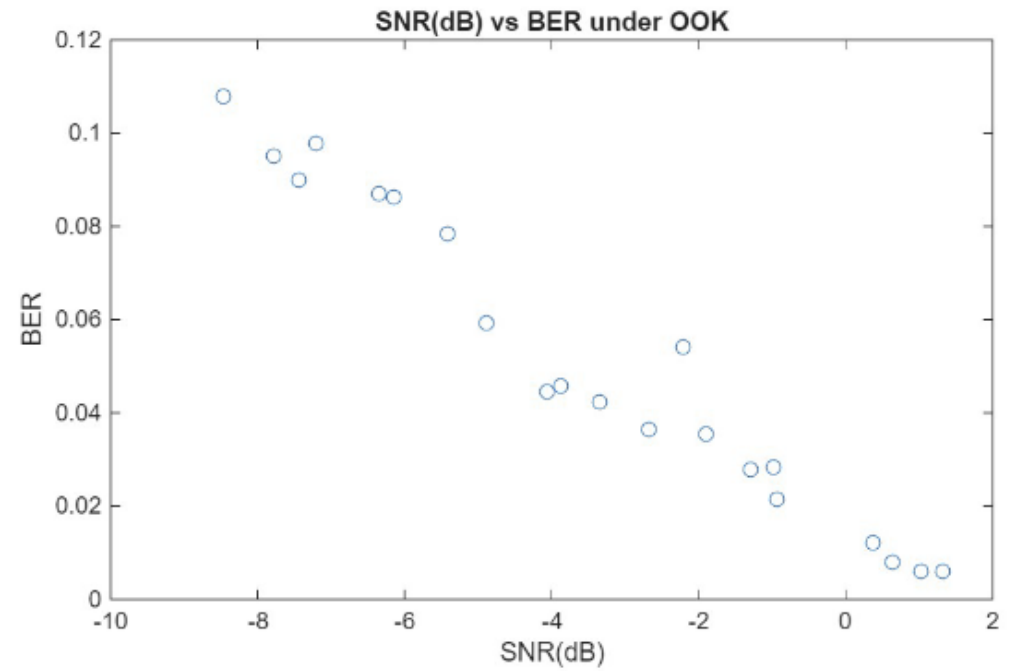
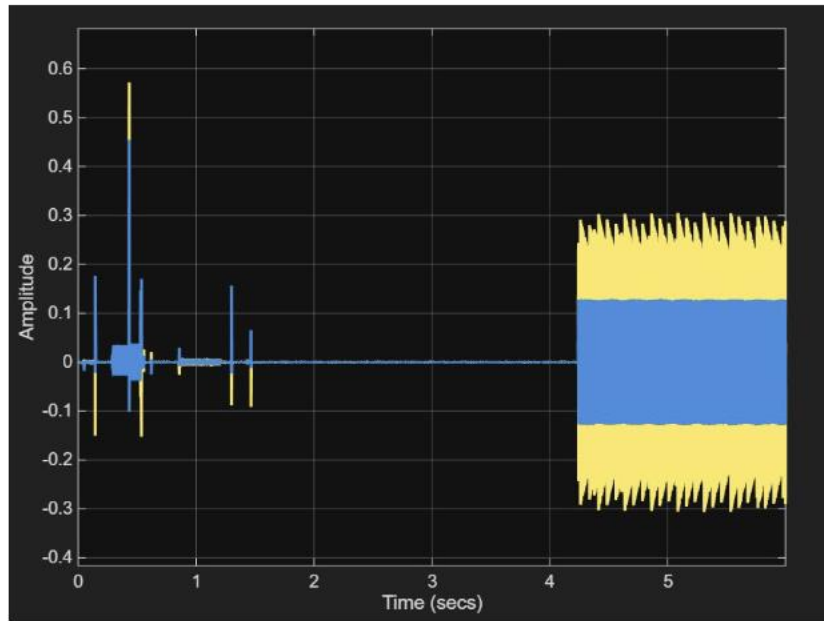
Hardware Pluto+ SDR



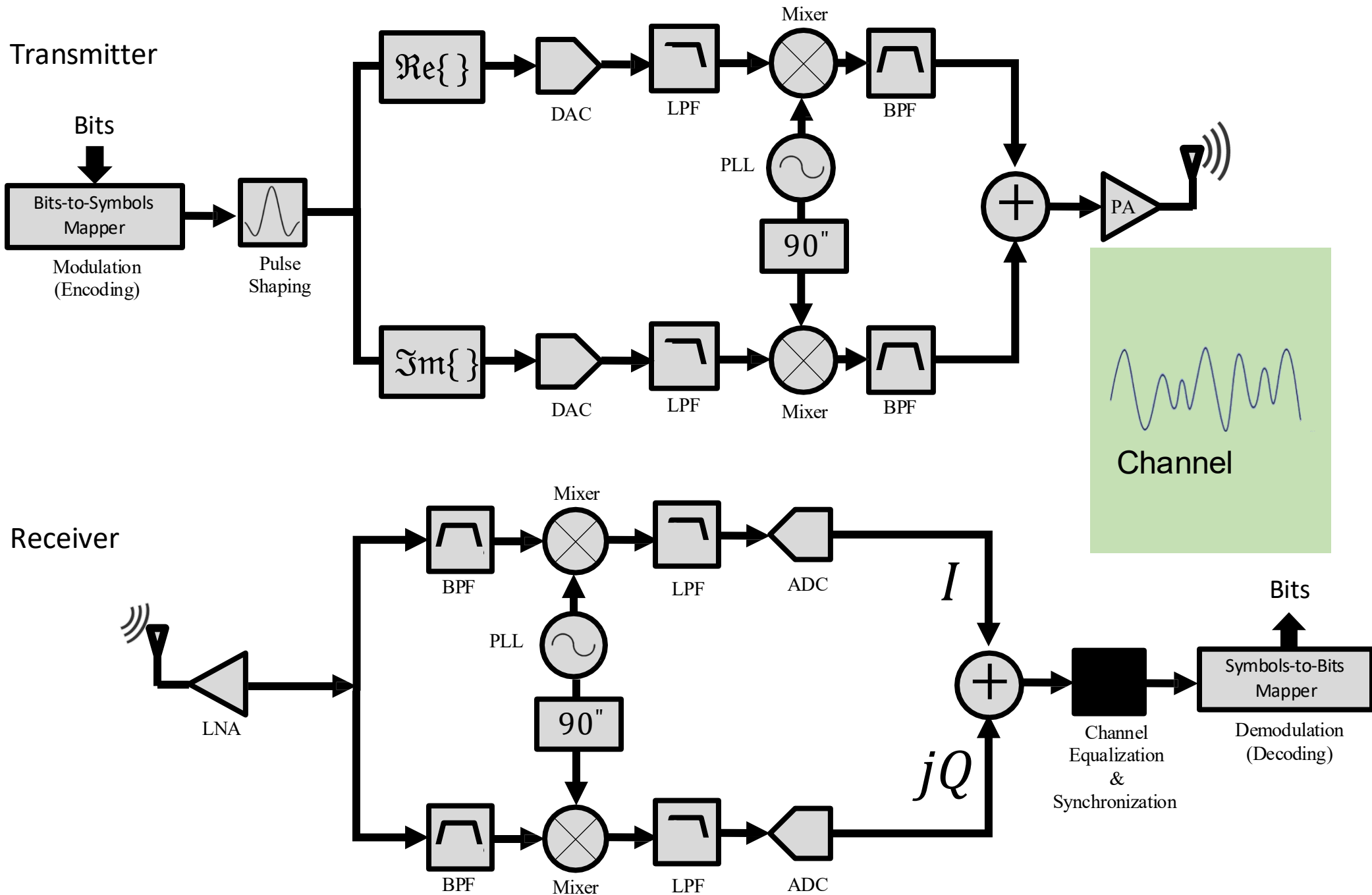
Hardware Pluto+ SDR



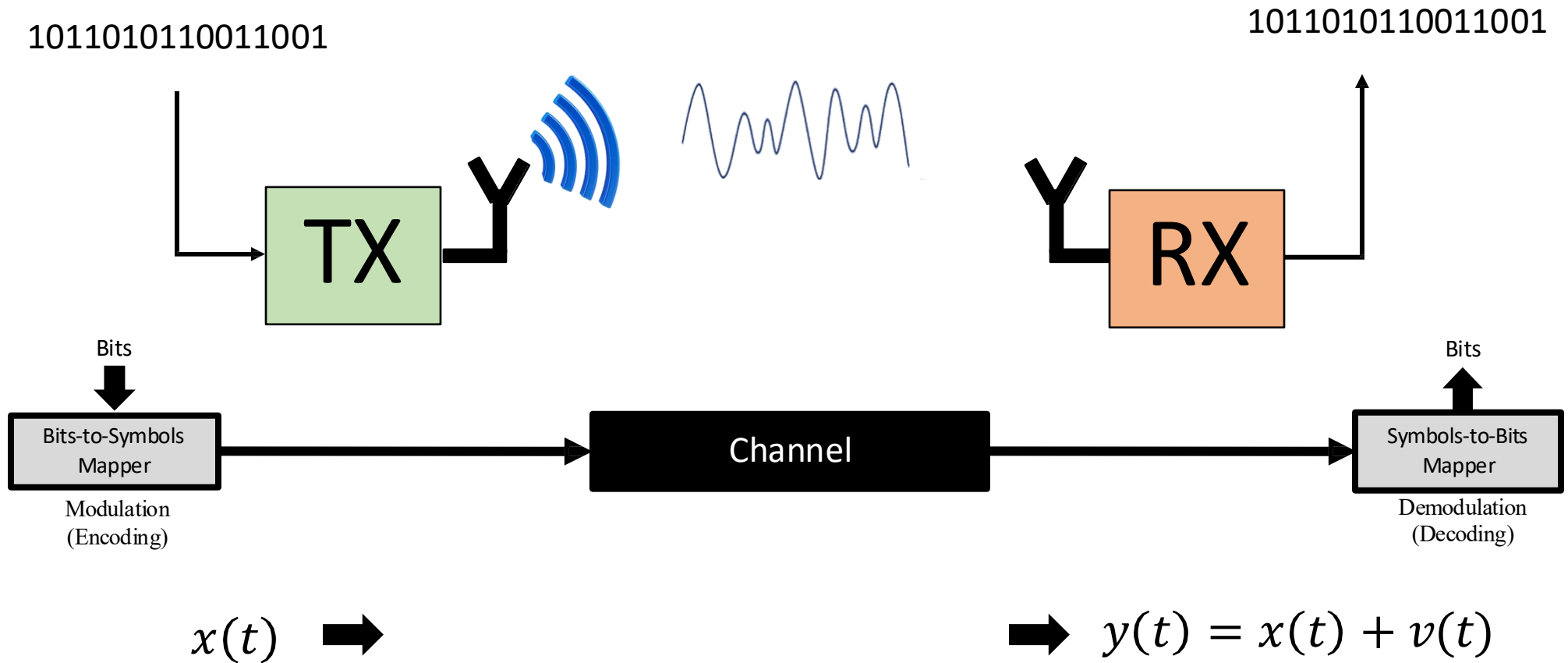
Hardware Pluto+ SDR



Wireless Communication System



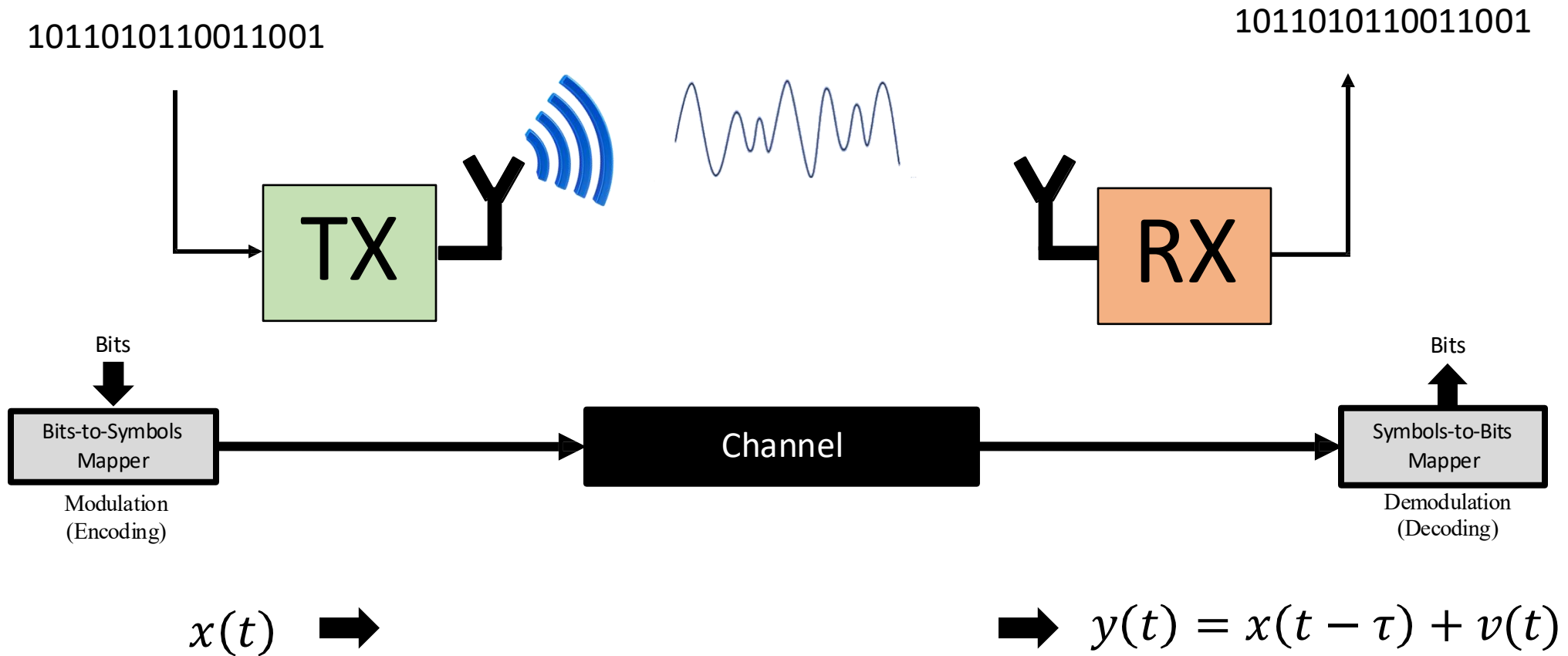
The Channel



Channel adds noise (AWGN)!

$$v(t) \sim N(0, \sigma^2)$$

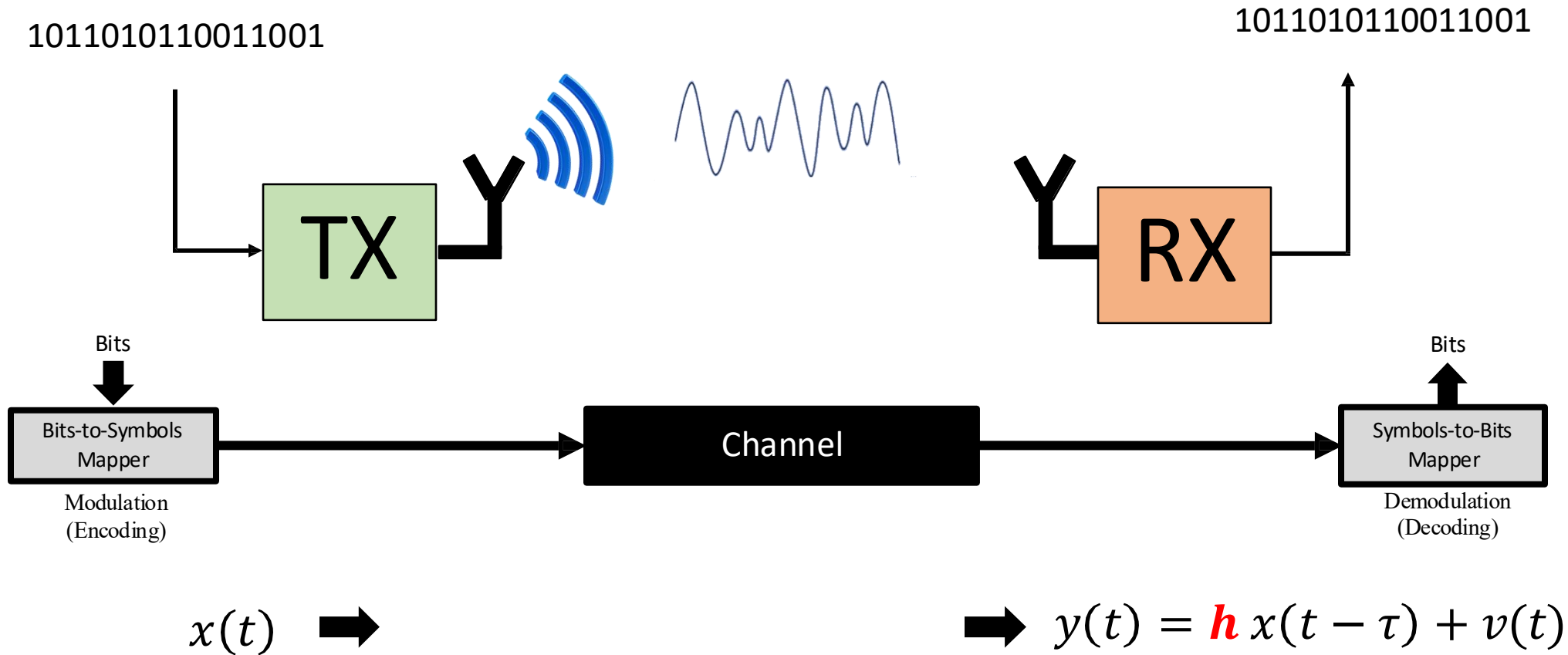
The Channel



Channel delays the signal!

$$\tau = \frac{d}{c}$$

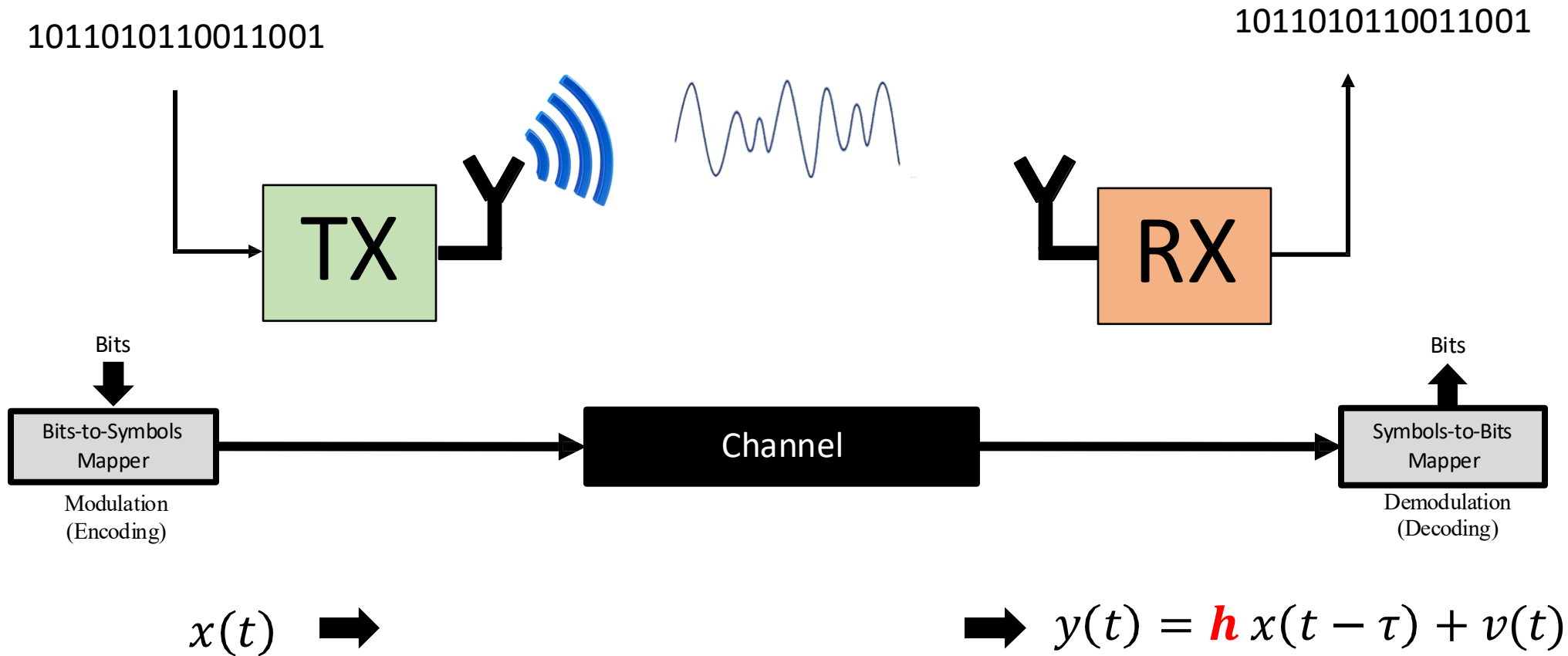
The Channel



Channel attenuates the signal (Pathloss)

$$P_{\text{RX}} = G_{\text{TX}} G_{\text{RX}} \frac{\lambda^2}{(4\pi d)^2} P_{\text{TX}} \Rightarrow |h| \propto \frac{\lambda}{d}$$

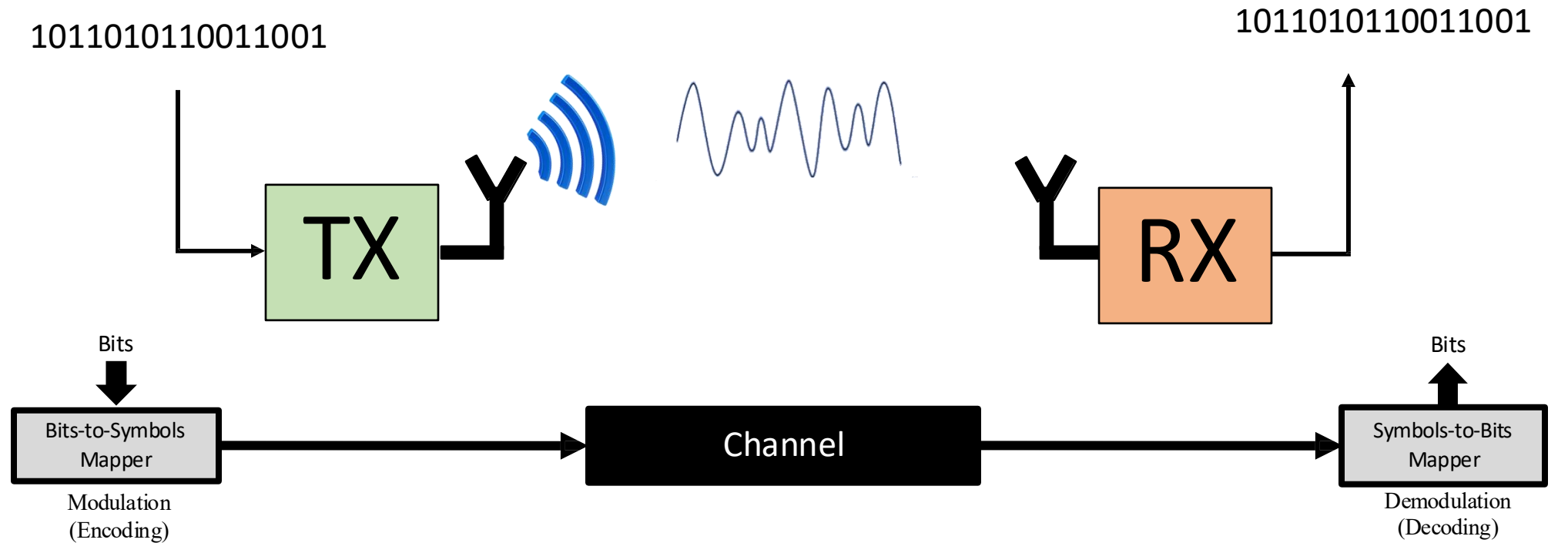
The Channel



Channel rotates the signal (Adds Phase)

$$h \propto \frac{\lambda}{d} e^{j\phi}$$

The Channel



$$x(t) \Rightarrow$$

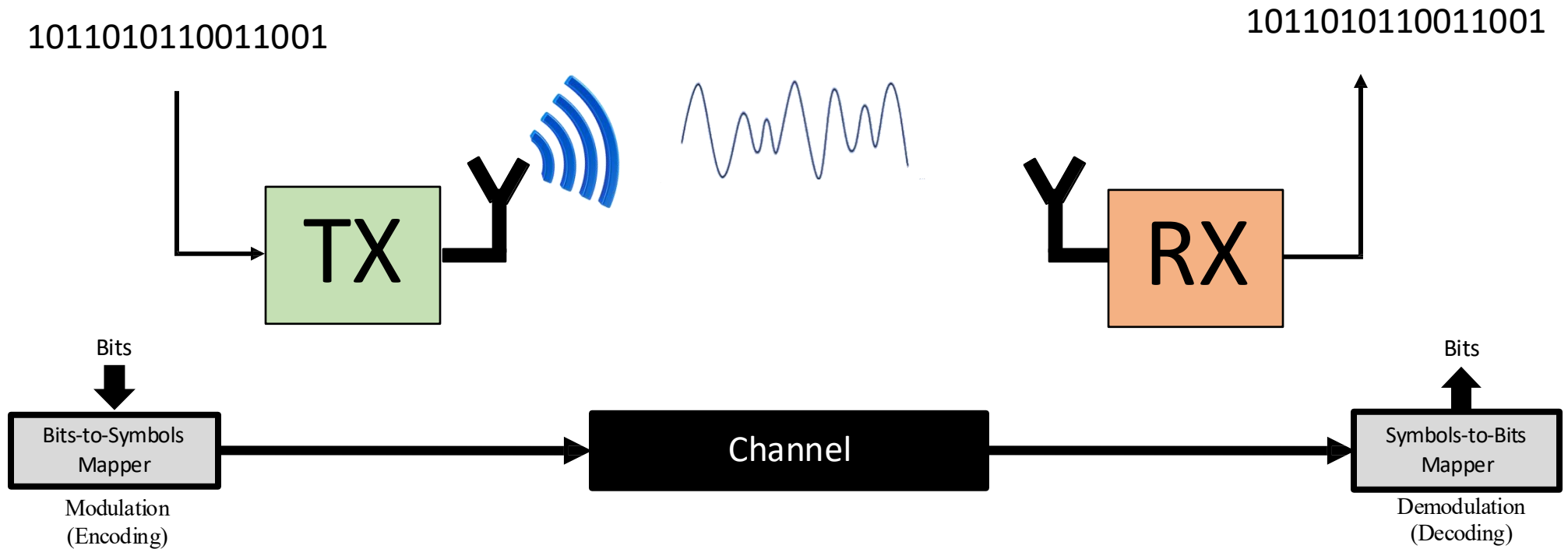
$$\Rightarrow y(t) = \mathbf{h} x(t - \tau) + v(t)$$

$$x(t) \times e^{-j2\pi f_c t} \Rightarrow |h| x(t - \tau) e^{-j2\pi f_c (t - \tau)} \Rightarrow \times e^{j2\pi f_c t} \Rightarrow |h| x(t - \tau) e^{j2\pi f_c t}$$

$$\mathbf{h} \propto \frac{\lambda}{d} e^{j\phi} \rightarrow \phi = 2\pi f_c \tau = 2\pi \frac{c}{\lambda} \frac{d}{c} = 2\pi \frac{d}{\lambda} \rightarrow \mathbf{h} \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

The Channel

The Channel



$$x(t) \Rightarrow$$

$$\Rightarrow y(t) = \mathbf{h} x(t - \tau) + v(t)$$

Channel:

- Adds Noise
- Delays the Signal
- Attenuates the Signal
- Rotates the Phase of the Signal

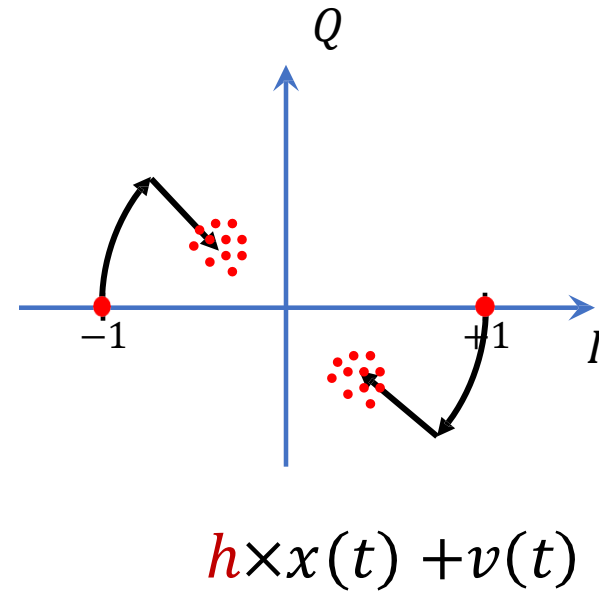
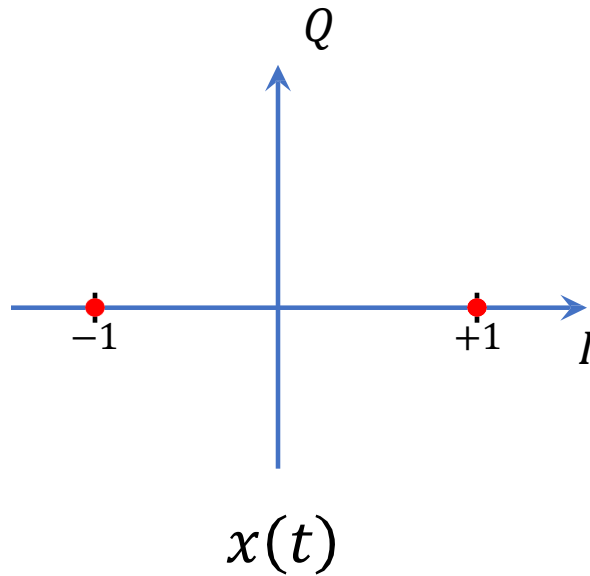
$$h \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

The Channel

Consider BPSK Modulation.

$$0 \rightarrow -1$$

$$1 \rightarrow +1$$

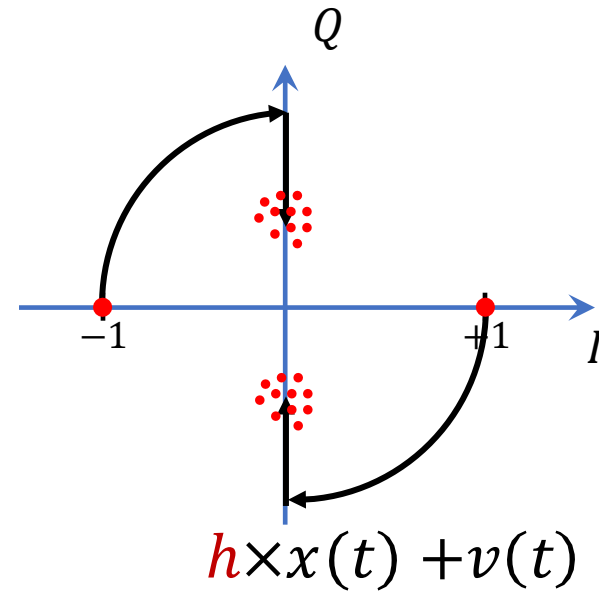
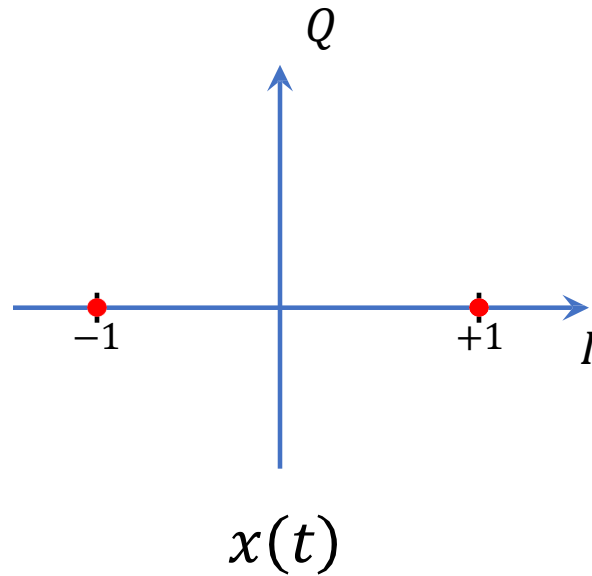


The Channel

Consider BPSK Modulation.

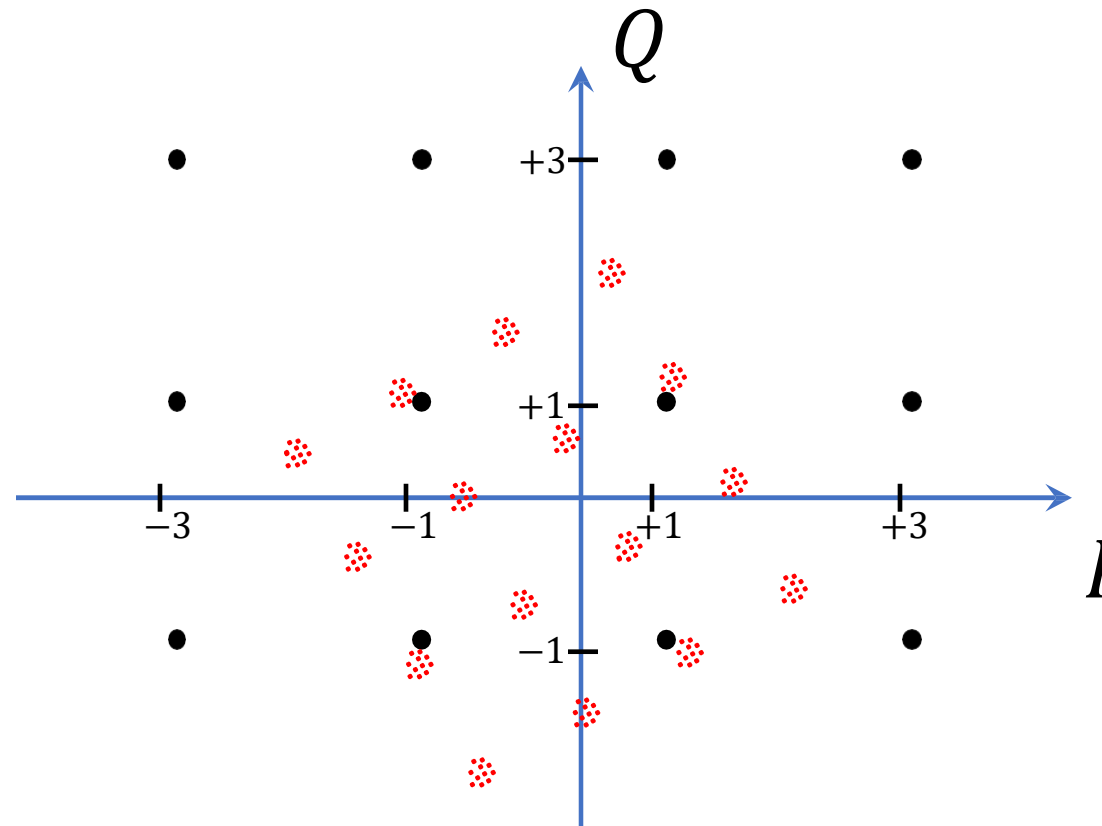
$$0 \rightarrow -1$$

$$1 \rightarrow +1$$



The Channel

Consider QAM Modulation



Demodulating correctly requires COHERENCE!
i.e., Need to estimate & correct for the channel h



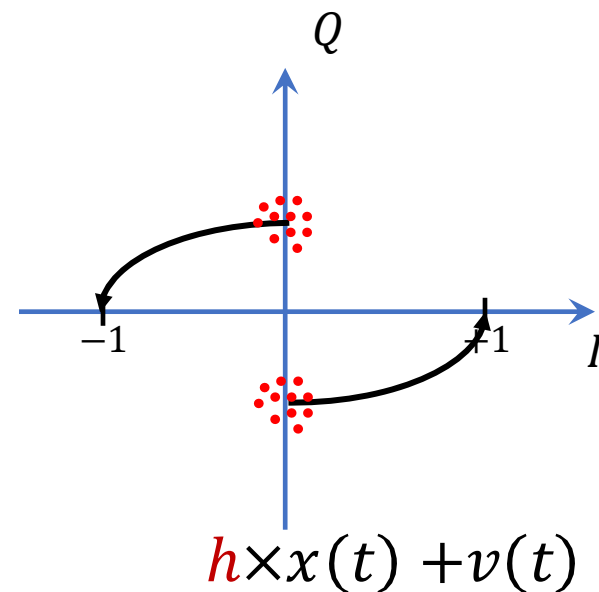
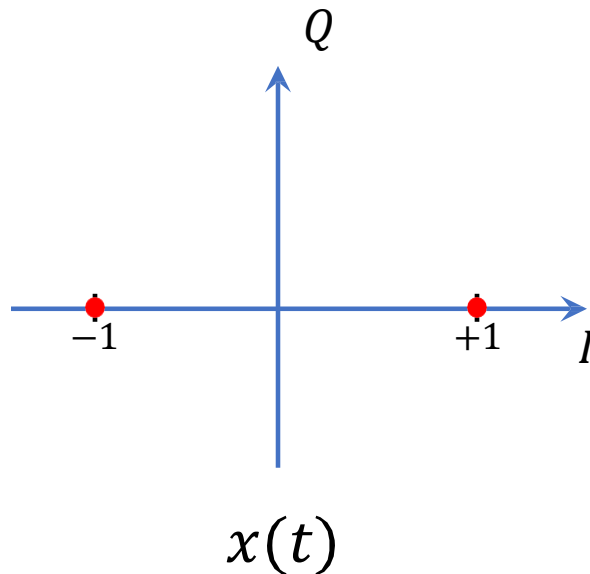
Channel Estimation & Correction

Channel Estimation & Correction

Consider BPSK Modulation.

$$0 \rightarrow -1$$

$$1 \rightarrow +1$$



Send Training Sequence (Preamble Bits): Known Bits

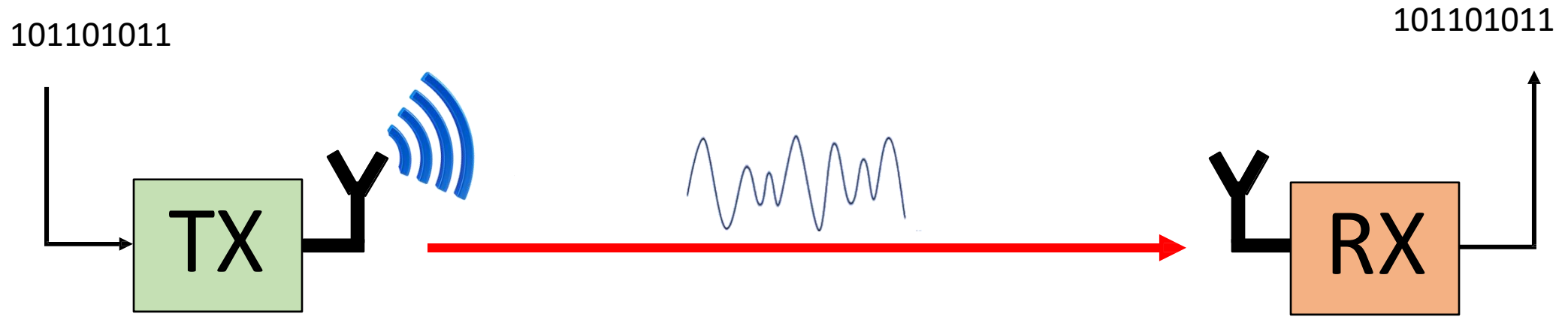
$$\begin{aligned} x(0) = 1 &\longrightarrow y(0) = h + v(0) \\ x(1) = 1 &\longrightarrow y(1) = h + v(1) \\ x(2) = -1 &\longrightarrow y(2) = -h + v(2) \end{aligned}$$

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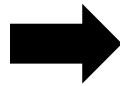
$$\text{Estimate channel: } \tilde{h} = \frac{1}{K} \sum_{k=1}^K \frac{y(k)}{x(k)}$$

$$\text{Correct channel: } \tilde{x}(t) = \frac{y(t)}{\tilde{h}(t)}$$

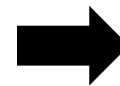
The Channel



$x(t)$



Channel

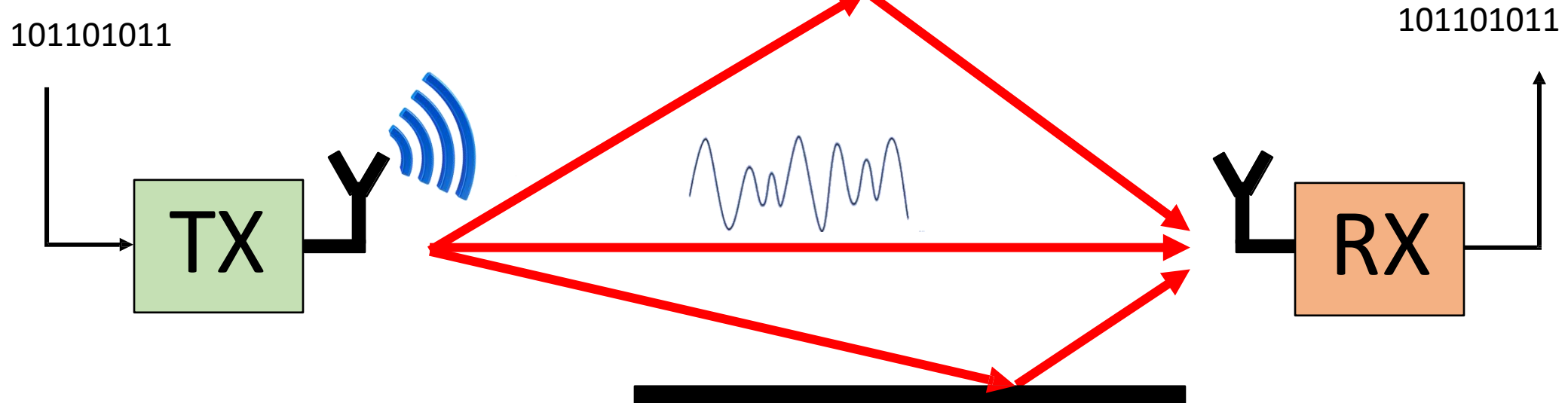


$y(t) = \mathbf{h} x(t - \tau) + v(t)$

$$h \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

Assumes single path!

Multipath Channel



Multipath Propagation: radio signal reflects off objects ground, arriving at destination at slightly different times

$$y(t) = \alpha_1 e^{j\phi_1} x(t - \tau_1) + \alpha_2 e^{j\phi_2} x(t - \tau_2) + \alpha_3 e^{j\phi_3} x(t - \tau_3) \dots$$

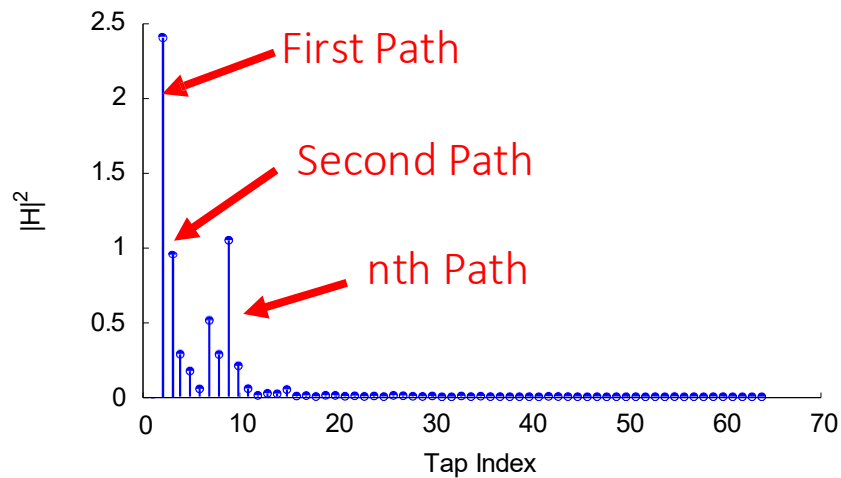
$$y(t) = \sum_k \alpha_k e^{j\phi_k} x(t - \tau_k) = \sum_k h_k(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

$h(t)$ is channel impulse response.

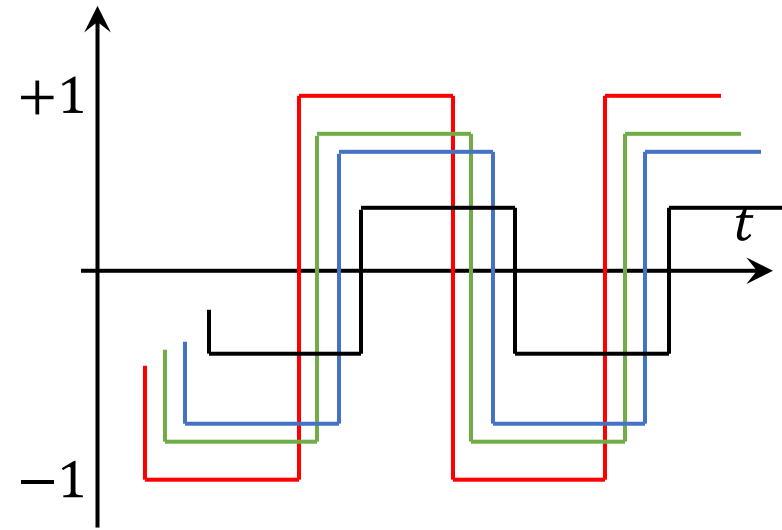
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$



Multi-tap Channel



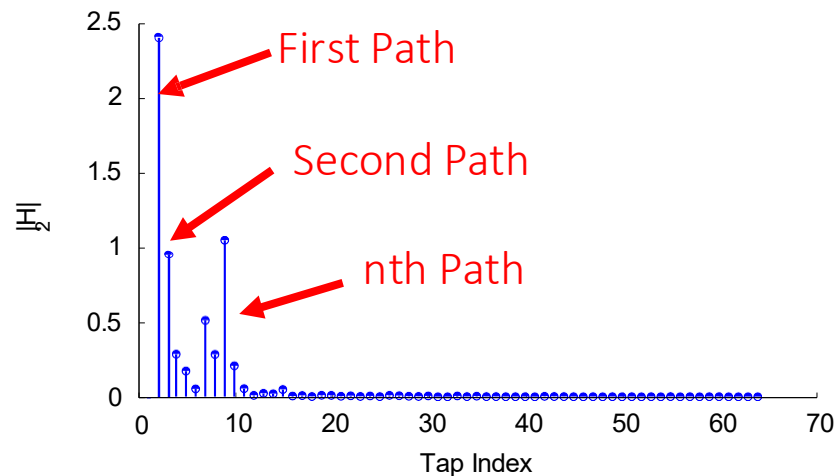
ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.

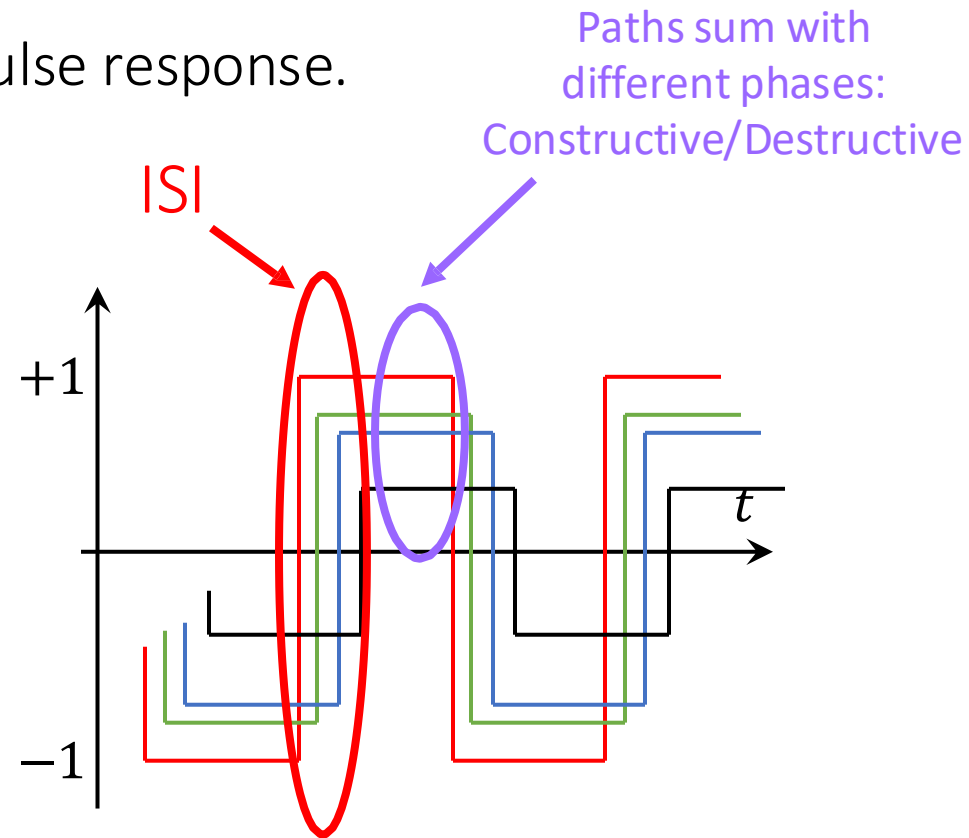
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Multi-tap Channel



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Channel Fading

Symbols arriving along different paths sum up destructively

Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

Channel Fading: Symbols arriving along different paths sum up destructively

Example 2 paths with distance $d_1 = 1m$, $d_2 = 1.06m$:

$$h = h_1 + h_2 = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} + \frac{\lambda}{d_2} e^{j2\pi d_2/\lambda}$$

$$= \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} \left(1 + \frac{d_1}{d_2} e^{j2\pi(d_2-d_1)/\lambda} \right) \quad \frac{d_1}{d_2} \approx 1$$

$$\text{if } \frac{d_2 - d_1}{\lambda} \approx \frac{1}{2} \rightarrow h = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} (1 + e^{j\pi}) = 0 \quad \text{Destructive Interference!}$$

Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

Channel Fading: Symbols arriving along different paths sum up destructively

Example 2 paths with distance $d_1 = 1m, d_2 = 1.06m$:

$$h = h_1 + h_2 = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} + \frac{\lambda}{d_2} e^{j2\pi d_2/\lambda}$$

@ $f_1 = 2.5GHz$ ($\lambda = 12\text{ cm}$): $h = 0.12 e^{j\frac{2\pi}{3}} + 0.113 e^{j\frac{5\pi}{3}} \approx 0.006$

@ $f_2 = 5GHz$ ($\lambda = 6\text{ cm}$): $h = 0.06 e^{j\frac{5\pi}{3}} + 0.05 e^{j\frac{5\pi}{3}} \approx 0.116$

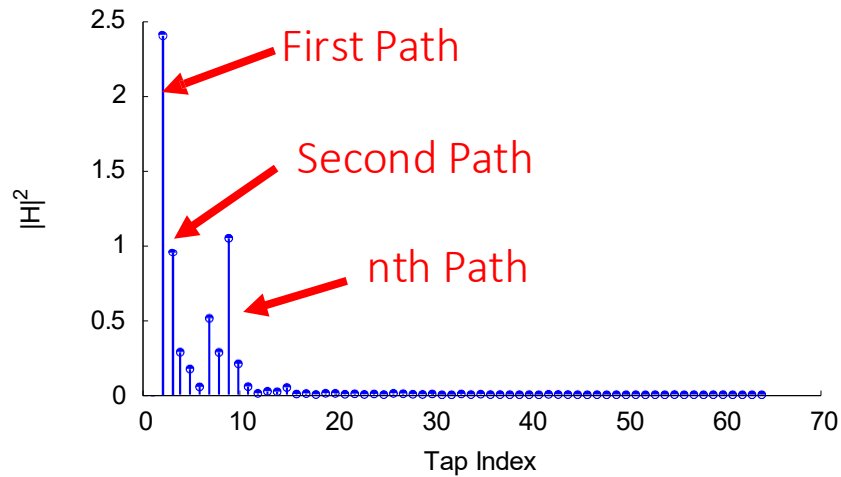
Frequency Selective Fading

17×
(24dB)

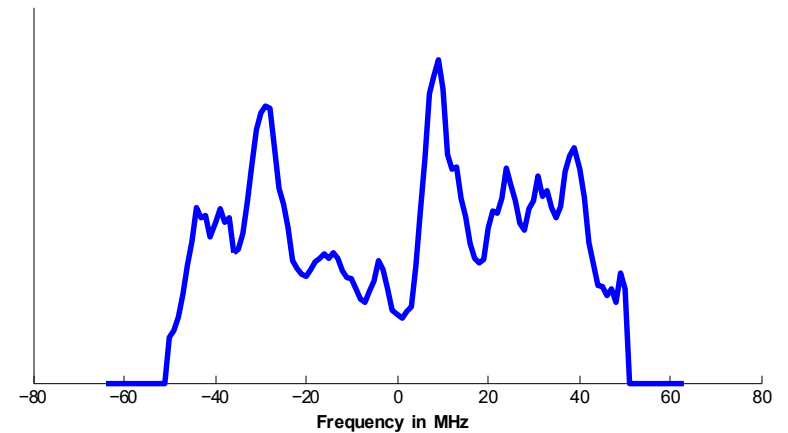
Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



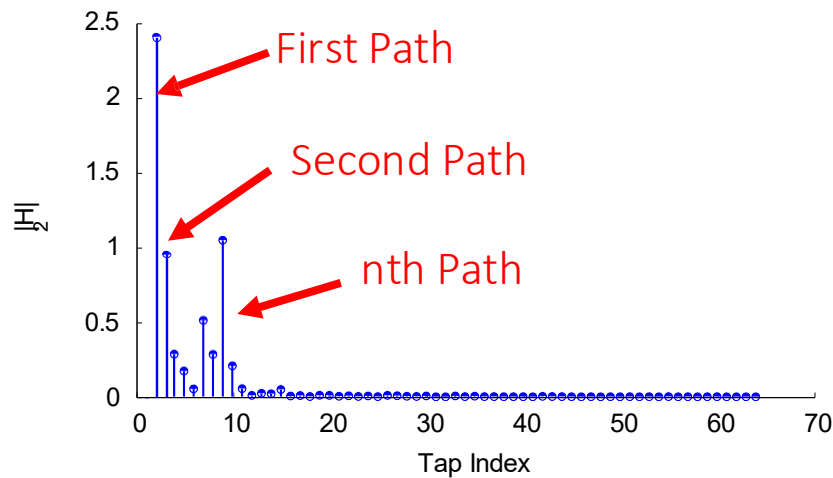
Multi-tap Channel



Multipath Channel

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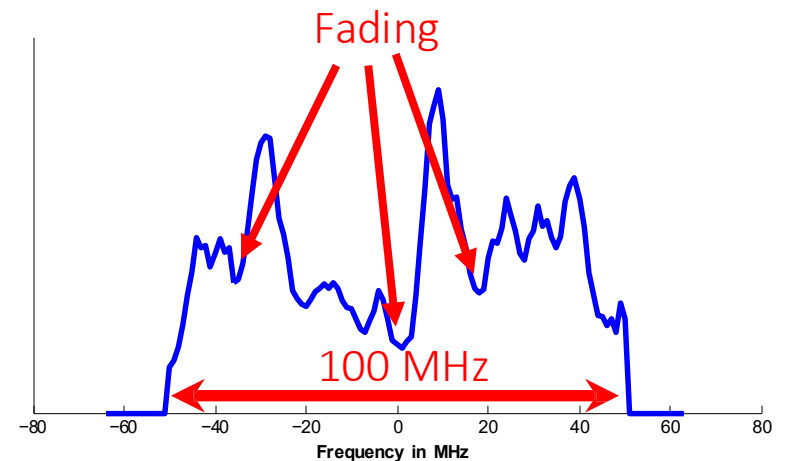


Multi-tap Channel



ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.



Frequency Selective Fading

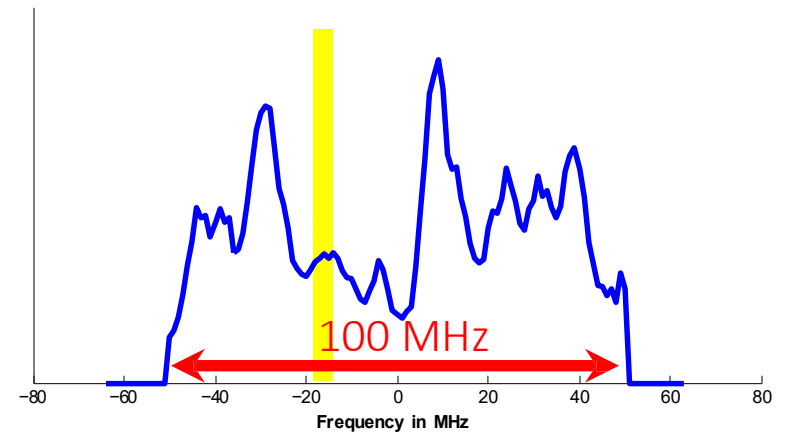
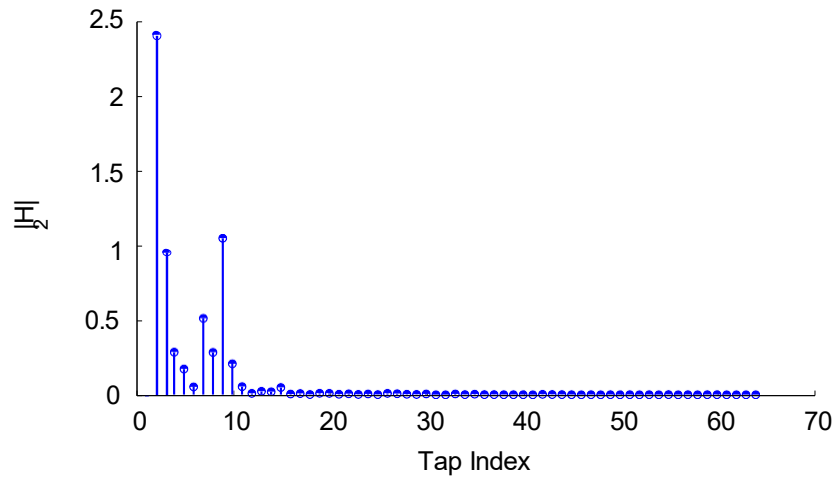
Symbols arriving along different paths sum up destructively

Problematic in
Wideband Channel!

Narrowband Channel

$h(t)$ is channel impulse response.

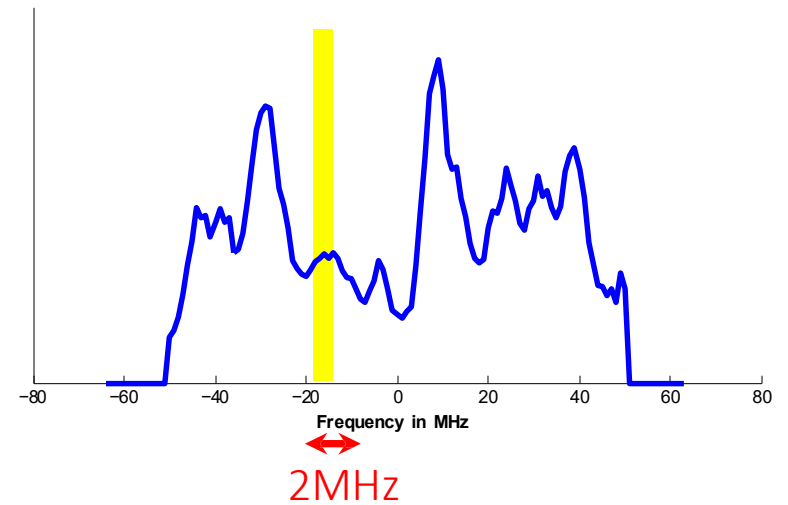
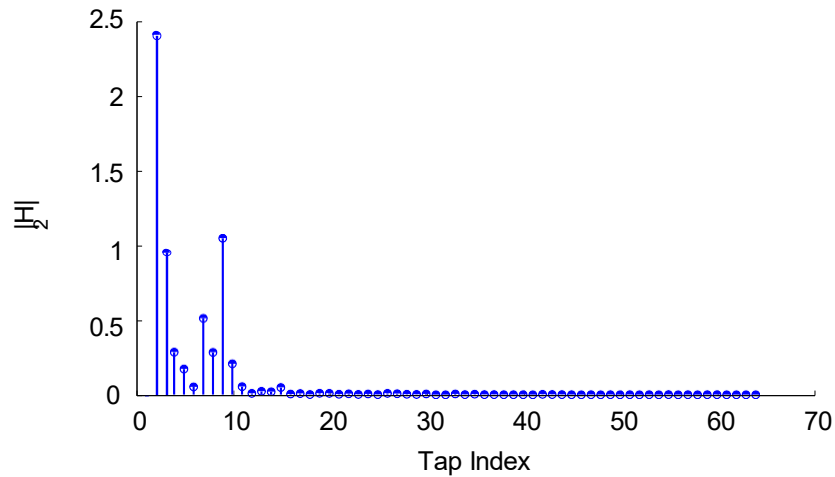
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Narrowband Channel

$h(t)$ is channel impulse response.

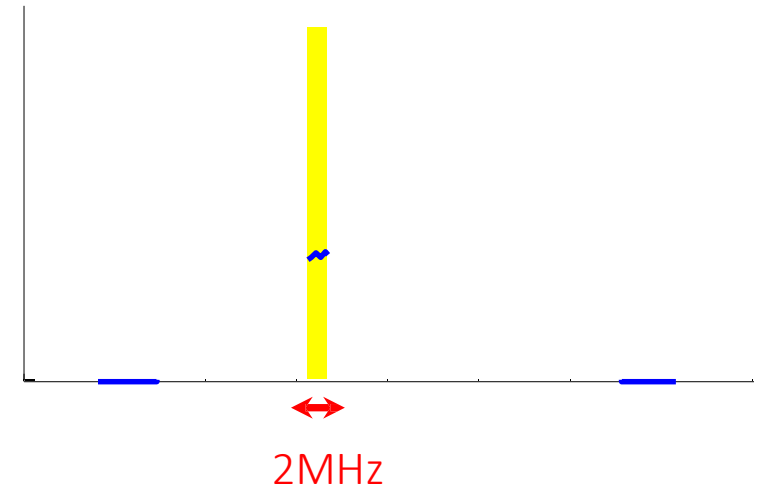
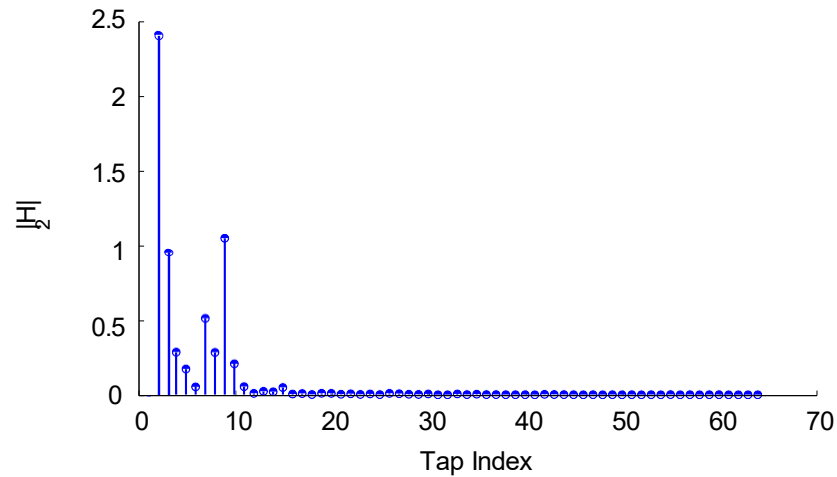
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Narrowband Channel

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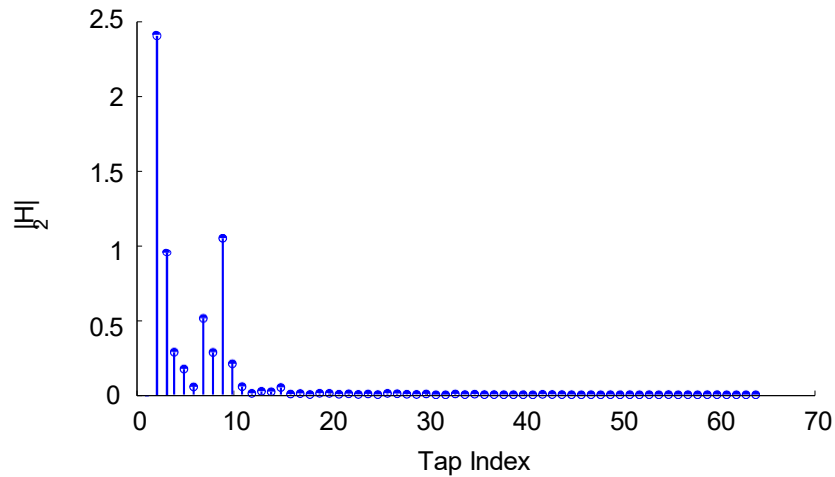
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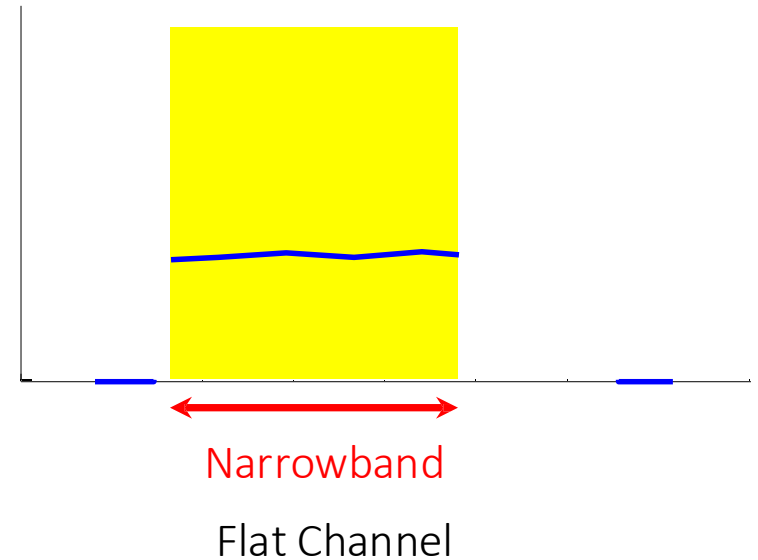
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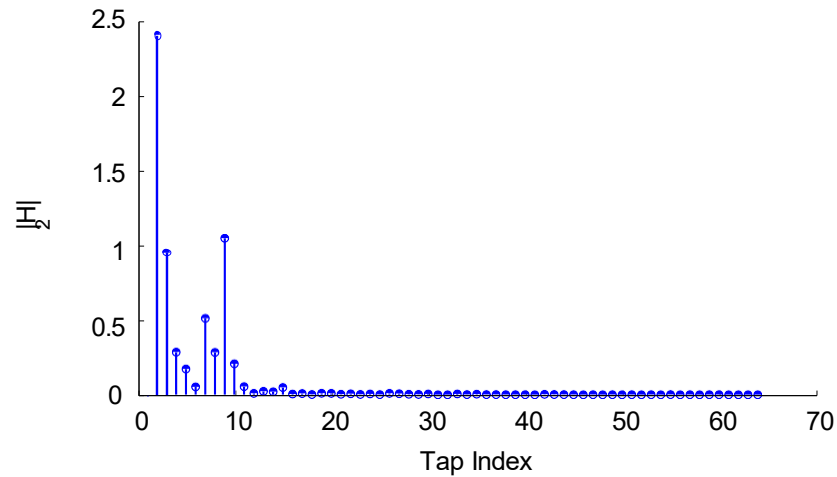
Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



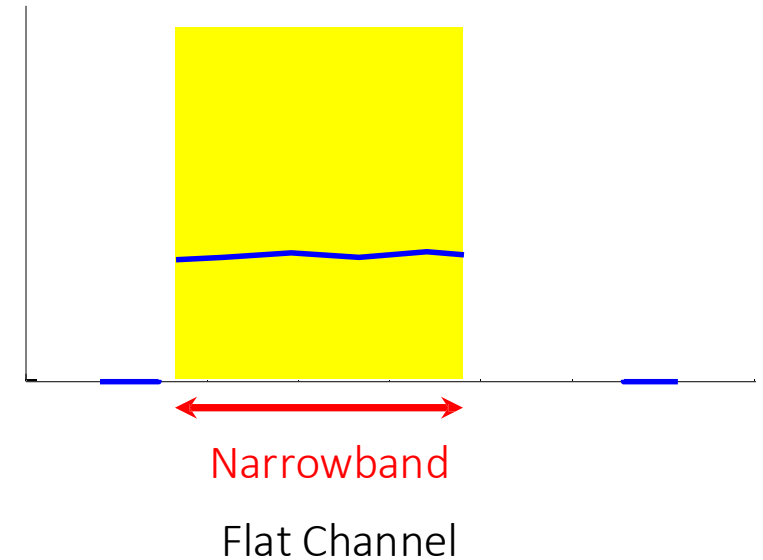
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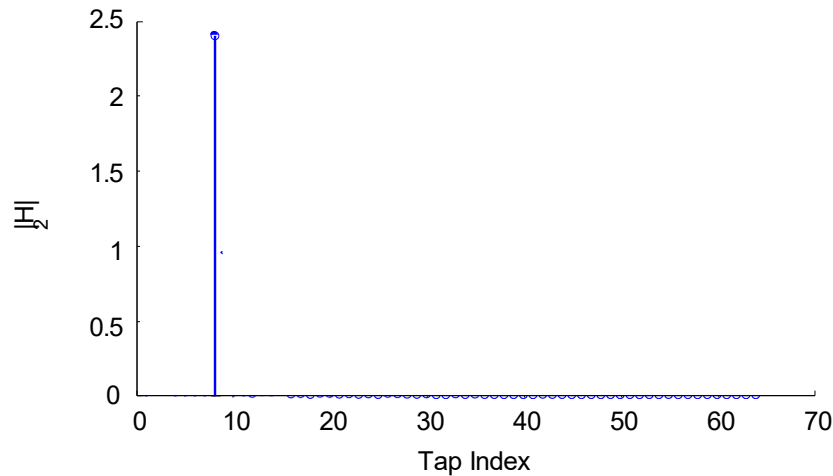
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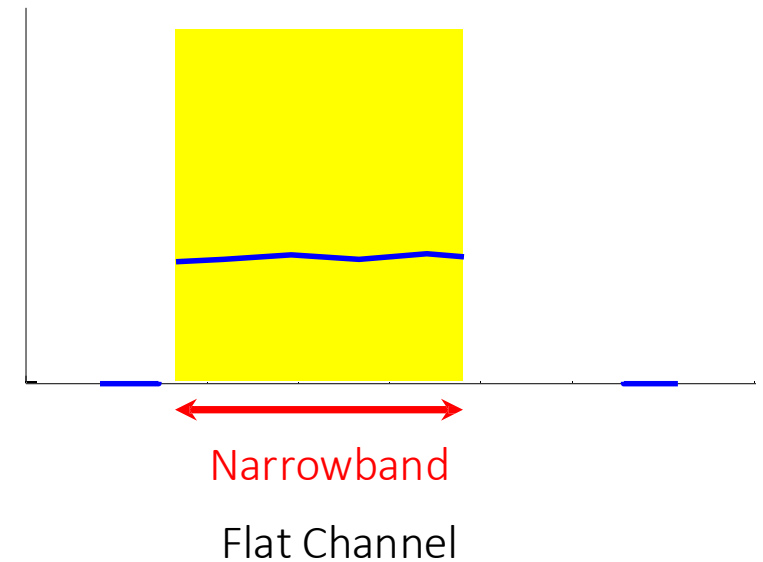
Narrowband Channel

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$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



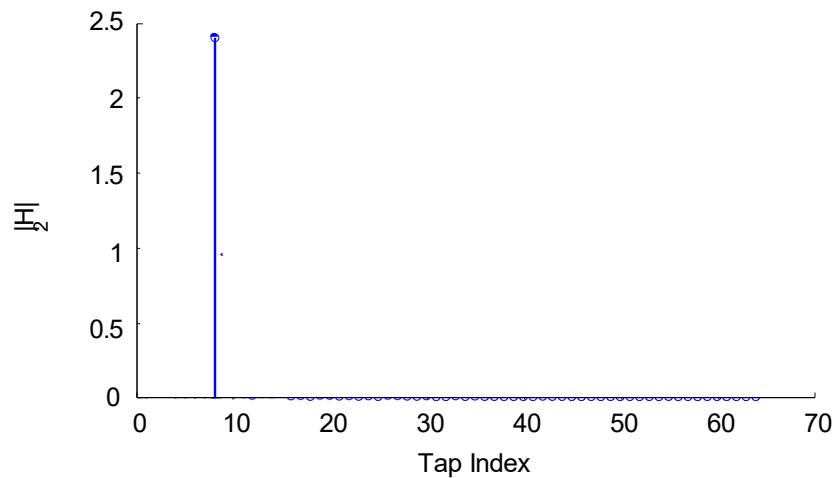
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx \sum_k h(\tau_k) x(t) = \left(\sum_k h(\tau_k) \right) x(t) = hx(t)$$

Narrowband Channel is Approximated by a Single Tap Channel

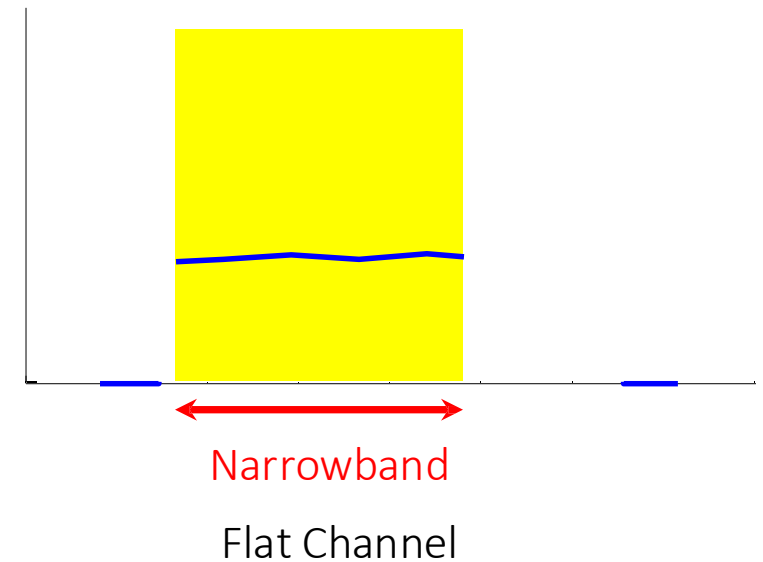
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t) = h x(t) \quad \Leftrightarrow \quad h X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



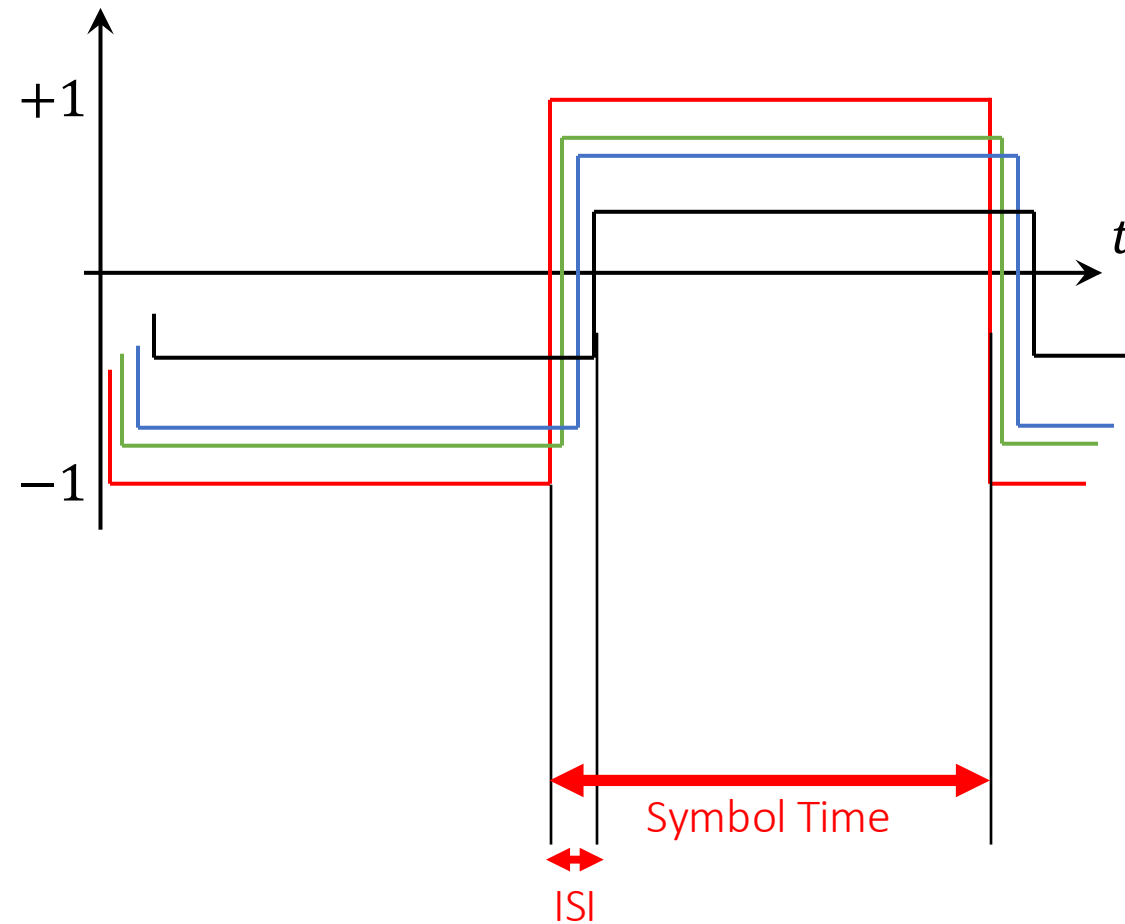
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Narrowband Channel is Approximated by a Single Tap Channel

Narrowband vs. Wideband Channel

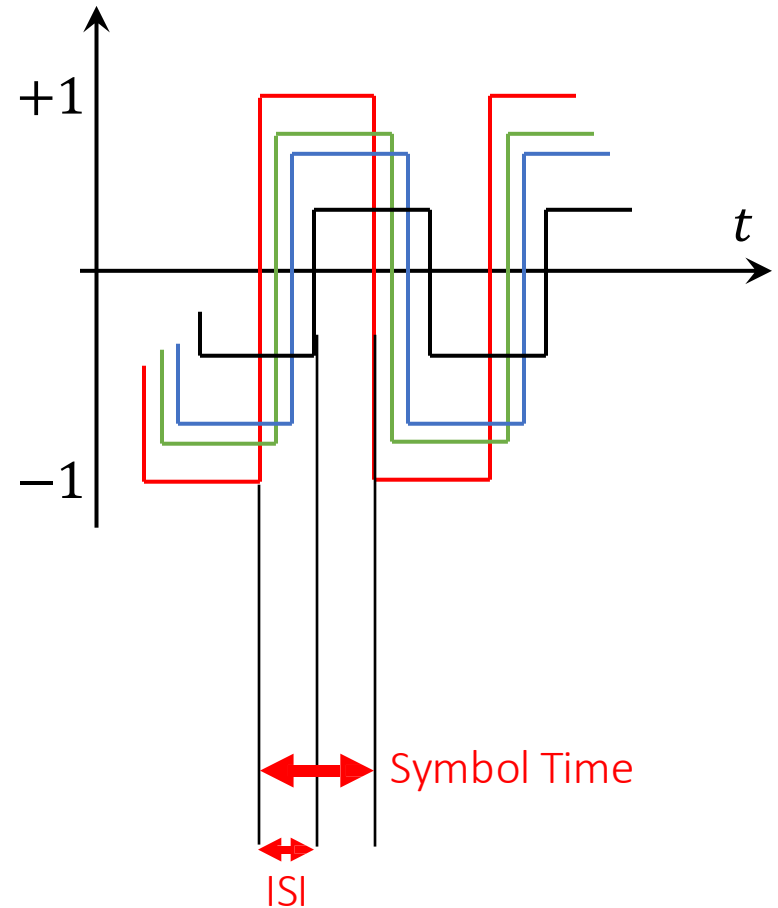
Narrowband Channel

$$h x(t)$$



Wideband Channel

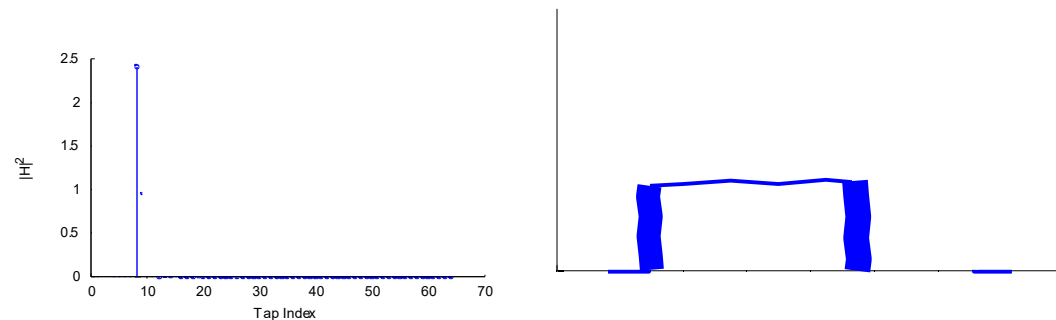
$$h(t) * x(t)$$



Narrowband vs. Wideband Channel

Narrowband Channel

$$h x(t) \iff h X(f)$$

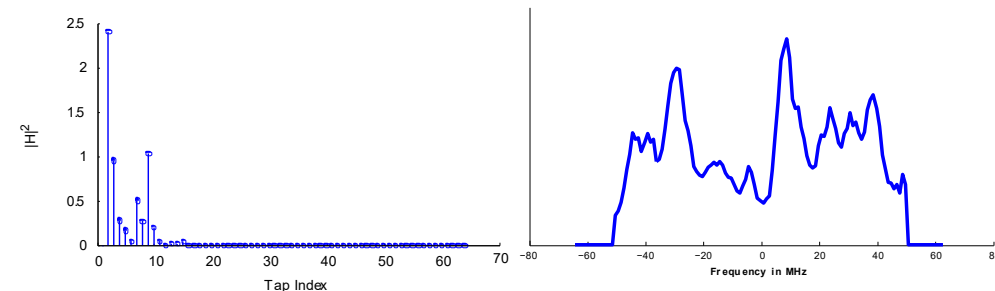


≈ Single Tap

≈ Flat Channel

Wideband Channel

$$h(t) * x(t) \iff H(f) X(f)$$



Multi-tap
Channel
(ISI)

Frequency
Selective Channel

Need to correct for ISI to
be able to decode
correctly!

Estimating and Correcting for multi-tap channel is hard
Simplify processing using: OFDM